TT-C-650C
October 22, 1968
SUPERSEDING
Fed. Spec. TT-C-650B
October 20, 1961

FEDERAL SPECIFICATION

CREOSOTE - COAL TAR SOLUTION

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers one type and five classes of creosote-coal tar solution derived entirely from tar produced by the carbonization of bituminous coal. It is intended for preservative treatment of piles, posts, lumber, structural timber, and cross, bridge, and switch ties (see 6.1).

1.2 Classification.

1.2.1 Type and classes. Creosote-coal tar solution shall be of one type and five classes as specified:

Class 1 - Sp. Gr. at 38°C. - 1.06-1.11
Class 2 - Sp. Gr. at 38°C. - 1.07-1.12
Class 3 - Sp. Gr. at 38°C. - 1.08-1.13
Class 4 - Sp. Gr. at 38°C. - 1.09-1.14
Class 5 - Sp. Gr. at 38°C. - Min.-1.090

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

Federal Specifications:

TT-W-571 - Wood Preservation: Treating Practices.
PPP-D-729 - Drums: Metal, 55-Gallon (For Shipment of Noncorrosive Material).
PPP-P-704 - Pails, Metal: (Shipping, Steel, 1 though 12 Gallon).

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized and Containerized Unit Loads 40" x 48" 4-Way (Partial) Pallet Skids, Runners, or Pallet-Type Base.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Laws and Regulations:

Code of Federal Regulations:

49 CFR 171-179 - Department of Transportation Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.
7 CFR 362 - Regulations For the Enforcement of the Federal Insecticide, Fungicide, and Rodenticide Act.

(Application for copies should be addressed to the U. S. Department of Agriculture, Agricultural Research Service, Pesticides Regulation Division, Washington, D. C. 20250.)

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified the issue in effect on date of invitation for bids or request for proposal shall apply.

American Wood Preservers' Association (AWPA) Standards:

A1-65 - Analysis of Creosote and Oil-Type Preservatives.
A4-56 - Sampling Wood preservatives.

(Application for copies should be addressed to the American Wood Preservers' Association, 1012 Fourteenth Street, N. W., Washington, D. C. 20005.)

3. REQUIREMENTS

3.1 Material. Creosote-coal tar solution shall be pure coal-tar product derived entirely from the carbonization of bituminous coal.

3.2 Properties. The properties of the creosote-coal tar solution shall comply with the requirements in table I for classes 1, 2, 3, 4, and 5, respectively.

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TABLE I. Requirements of creosote-coal tar solutions [1]

	Class 1		Class 2		Class 3		Class 4		Class 5	
	Not less than	Not more than	Not less than	Not more than	Not less than	Not more than	Not less than	Not more than	Not less than	Not more than
Composition: Coal-tar distillate, percent by volume.	80	--	70	--	60	--	50	--	60	75
Water, percent by volume.	--	3.0	--	3.0	--	3.0	--	3.0	--	3.0
Specific gravity at 38°C. compared with water at 15.5°C.	1.06	1.11	1.07	1.12	1.08	1.13	1.09	1.14	1.090	--
Material insoluble in benzol, percent by weight.	--	3.0	--	4.0	--	4.5	--	5.0	1.0	4.5
Coke residue, percent by weight.	--	6.0	--	8.0	--	10.0	--	12.0	4.0	8.0
Distillation: The distillate percent by weight on the waterfree basis shall be within the following limits:										
Up to 210°C.	--	5	--	5	--	5	--	5	--	2
Up to 235°C.	2	15	2	15	2	15	2	15	2	15
Up to 270°C.	--	--	--	--	--	--	--	--	20	35
Up to 315°C.	36	--	34	--	32	--	30	--	35	50
Up to 355°C.	60	--	56	--	52	--	48	--	55	70
Specific gravities of fractions: Fraction 235° to 315°C. at 38°C. compared with water at 15.5°C.	1.027	--	1.027	--	1.027	--	1.027	--	1.030	--
Fraction 315° to 355°C. at 38°C. compared with water at 15.5°C.	1.095	--	1.095	--	1.095	--	1.095	--	1.110	--
Fraction above 355°C. (distillation residue) at 38°C. compared with water at 15.5°C.									1.185	

1 These requirements apply to solutions in use at treating plants and are not subject to any exception or tolerances for changes taking place due to treating operations.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling. Sampling shall be conducted in accordance with AWPA Standard A4-56. The treating solution shall be thoroughly agitated after which a 1-quart sample shall be taken for inspection and test purposes. The contractor shall supply such samples and containers as may be required at no cost to the Government.

4.3 Test methods.

4.3.1 Properties. Tests to determine the properties of creosote-coal tar solution (see 3.2) shall be made in accordance with AWPA Standard A1-65.

4.3.2 Examination of preparation for delivery. An examination shall be made to determine that the packaging, packing palletization (when required) and marking comply with the requirements of section 5. Defects shall be scored as specified in table II. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 with an AQL of 4.0 expressed in terms of defects per hundred units.

TABLE II. Examination of preparation for delivery

Examine	Defect
Containers	Not as specified.
Contents	More or less than required.
Labeling and Marking	Omitted; incorrect; illegible; improper size, location, sequence or method of application.
Materials	Component missing or damaged.
Workmanship	Bulging or distortion of containers.
Palletization (when required)	Not as specified.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C as specified (see 6.2).

5.1.1 Levels A and B. Unless otherwise specified in the contract or order, the solution shall be packaged in five-gallon pails conforming to PPP-P-704, type I, class 1 or 2, except commercial exterior coatings are acceptable for level B. Pails shall be furnished with pour spouts.

5.1.2 Level C. The creosote solution shall be packaged to provide adequate protection against corrosion, deterioration, and damage from the supply source to the first receiving activity for immediate use.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Levels A and B.

5.2.1.1 Five-gallon pails. Unless otherwise specified in contract or order (see 6.2), no overpacking of filled 5-gallon pails shall be required. When specified, filled 5-gallon pails shall be palletized in accordance with the requirements for load type III of MIL-STD-147.

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5.2.1.2 Fifty-five gallon drums. Fifty-five gallons of the solution shall be packed in nominal 55-gallon drums conforming to type I or II of PPP-D-729, as specified (see 6.2).

5.2.1.3 Bulk. Tank-car shipments of the solution shall be prepared in accordance with the Department of Transportation Rules and Regulations.

5.2.2 Creosote solution in quantities as specified (see 6.2), packaged as specified in 5.1, shall be packed in containers to assure carrier acceptance and safe arrival at destination in accordance with the Department of Transportation Rules and Regulations.

5.3 Labeling and Marking. The labeling and marking shall comply with the Federal Insecticide, Fungicide, and Rodenticide Act and the Department of Transportation Rules and Regulations. In addition to any special marking required by the procument document, marking of the interior packages and exterior shipping containers shall be in accordance with Fed. Std. No. 123 for civil agencies MIL-STD-129 for military agencies as applicable (see 6.2).

5.3.1 Labeling. All shipping containers shall be labeled in accordance with applicable Department of Transportation Rules and Regulations. The labeling shall be registered under the Federal Insecticide, Fungicide, and Rodenticide Act, as amended, in compliance with its requirements. Unless otherwise specified, each individual container shall be plainly marked with the following information:

Creosote-Coal Tar Solution.
Federal Specification TT-C-650c.
Class.
Quantity.
Gross weight.
Name of manufacturer (name of contractor if not the same).
Government order number.

CAUTION:

May cause skin irritation.
Avoid prolonged or repeated contact with skin.
Avoid prolonged breathing of vapor.
Use only with adequate ventilation.

6. NOTES

6.1 Intended use. The intended use of this material (see Federal Specification TT-W-571g) is in general as follows:

Classes 1 and 2 - For land piles, posts, lumber, structural timber, and bridge ties.
Classes 3 and 4 - For cross ties and switch ties.
Class 5 - For piles, lumber, and structural timber--all used in coastal waters.

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6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and data of this specification.
(b) Class required (see 1.2.1 and 6.1).
(c) Administrative provision for inspection records (see 4.1).
(d) Size of container required (see 5.1.1 and 5.2.1.2).
(e) Selection of applicable level of packaging and packing required (see 5.1 and 5.2).
(f) Quantities required (see 5.2.2).
(g) Marking documents required (see 5.3).
(h) Palletization, if required (see 5.2.1.1).

MILITARY INTEREST:

Custodians:

Army - ME
Navy - YD
Air Force - 68

Preparing activity

AGR-FS

Review activities:

Army - CE, MD, MU
Navy - YD
Air Force - 68

User activities:

Army - AT, SM
Navy - CG

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